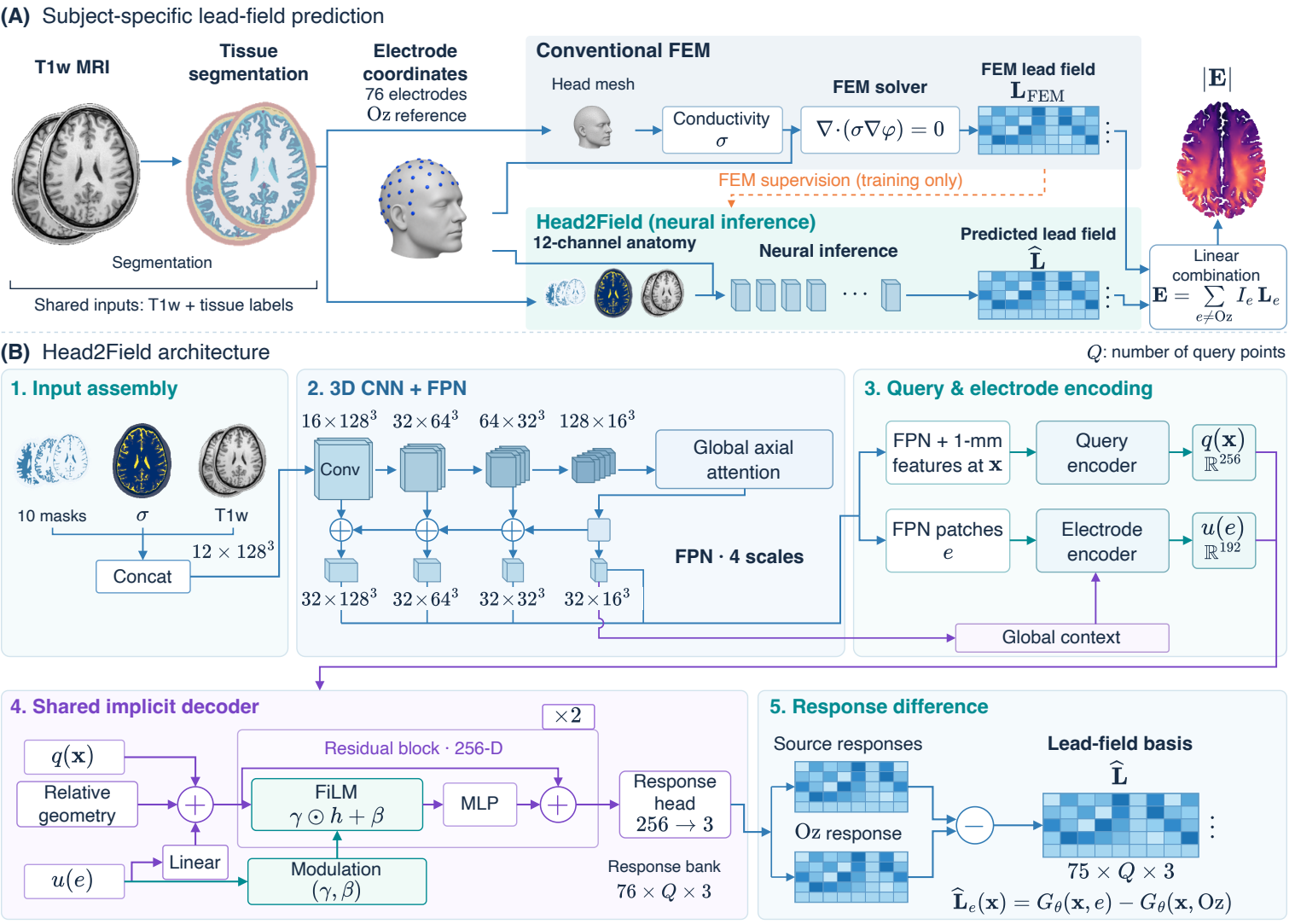

(A) Subject-specific lead-field prediction
T1w MRI
Tissue segmentation
Segmentation
Electrode coordinates
76 electrodes
Oz reference
Shared inputs: T1w + tissue labels
Conventional FEM
Head mesh
Conductivity σ
FEM solver
$\nabla\cdot(\sigma\nabla\varphi)=0$
FEM lead field $\mathbf{L}_{\mathrm{FEM}}$
FEM supervision (training only)
Head2Field (neural inference)
12-channel anatomy
Neural inference
Predicted lead field $\widehat{\mathbf{L}}$
$|\mathbf{E}|$
Linear combination $\mathbf{E}=\sum_{e\neq \mathrm{Oz}} I_e\,\mathbf{L}_e$
(B) Head2Field architecture
Q: number of query points
1. Input assembly
10 masks
σ
T1w
Concat
12×128^3
2. 3D CNN + FPN
16×128^3
32×64^3
64×32^3
128×16^3
Conv
Global axial attention
FPN · 4 scales
32×128^3
32×64^3
32×32^3
32×16^3
3. Query & electrode encoding
FPN + 1-mm features at $\mathbf{x}$
Query encoder
$q(\mathbf{x})$ $\mathbb{R}^{256}$
FPN patches e
Electrode encoder
$u(e)$ $\mathbb{R}^{192}$
Global context
4. Shared implicit decoder
$\times2$
Residual block · 256-D
$q(\mathbf{x})$
Relative geometry
FiLM $\gamma\odot h+\beta$
MLP
Response head $256\to3$
$u(e)$
Linear
Modulation (γ,β)
Response bank $76\times Q\times3$
5. Response difference
Source responses
Oz response
Lead-field basis $\widehat{\mathbf{L}}$
$75\times Q\times3$
$\widehat{\mathbf{L}}_e(\mathbf{x})=G_\theta(\mathbf{x},e)-G_\theta(\mathbf{x},\mathrm{Oz})$